

Sami Ahmed Mohammed

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EXPERIENCE

University of Central Florida Graduate Teaching Assistant - Support STA 2023: Statistical Methods I by grading coursework, assignments, and assessments. - Assist with preparing lecture materials and course resources. - Help students understand statistical concepts and resolve questions during office hours and class discussions.	Orlando, FL 08/2026 - Present
University of Central Florida Graduate Student Assistant - Supported faculty and students with course coordination and day-to-day academic operations. - Responded to student inquiries and coordinated communication between students and faculty. - Assisted with administrative and academic tasks while working closely with faculty and staff.	Orlando, FL 10/2025 - 08/2026
Dhaaps Edutech Data Science Intern - Designed a machine learning-based solution to improve loan approval accuracy by 20%. - Utilized Python for data analytics during a month-long internship, showcasing punctuality and problem-solving. - Automated data processing tasks using Python, boosting efficiency in educational tools.	Hyderabad, TG 05/2023 - 06/2023
SEO Content AI Machine Learning Engineer Intern - Expanded global audience reach by integrating multi-language support via Google Translate API. - Developed a Python and JavaScript Chrome extension, improving content quality and speed by 40%. - Led the integration of transformer models in AWS microservices, enhancing AI-driven content generation by 25%. - Deployed scalable applications using Docker, AWS ECS, and Fargate for reliable cloud-based solutions. - Implemented agile methodologies to improve team collaboration and streamline project management. - Enhanced content quality with BERT, RoBERTa, and DistilBERT for grammar correction, improving user trust.	Los Angeles, CA 04/2023 - 12/2023

PROJECTS

Chronic Kidney Disease Predictor (MERN Stack Random Forest Flask MongoDB Python, XGBoost) - Developed logistic regression model using Scikit-learn achieving 98% F1 score for kidney disease prediction. - Architected full-stack MERN application with Flask backend, creating intuitive clinical interface for healthcare professionals. - Implemented MongoDB storage for secure patient data management, enabling scalable healthcare data workflows.	[Link]
NutriPedia (TensorFlow, Inception-V3, GCP Storage, OpenAI GPT-3.5 API, SQL, MERN Stack) - Architected cloud-based nutrition platform integrating Inception-V3 CNN with GCP Storage for image processing and GPT-3.5 API for personalized dietary insights from SQL nutritional database. - Engineered full-stack data pipeline connecting computer vision classifications with NLP-powered contextual responses, creating a flywheel architecture for continuous model improvement through user feedback.	[Link]
Flight Fare Predictor (Random Forest, ExtraTrees, Flask, Scikit-learn, Feature Engineering) - Developed regression model using Scikit-learn and LSTM networks achieving dynamic flight price forecasting with temporal analysis. - Engineered feature extraction pipeline with Pandas identifying key price determinants, improving model accuracy from historical data. - Created data visualizations using Matplotlib/Seaborn for trend analysis and model performance interpretation for predictions.	[Link]
Beverage Management System (YOLOv5, Streamlit, SQLite, WebRTC, Computer Vision, PyTorch) - Implemented YOLOv5 model, achieving real-time beverage detection with 98% accuracy. - Developed a web application using Flask as the backend and Streamlit for the frontend, offering a seamless user experience. - Leveraged PostgreSQL for database management, ensuring robust and reliable storage of inventory data. - Incorporated Docker for deploying the application, ensuring consistency across development and production environments.	[Link]
Telco Customer Churn Pipeline (Python, MLflow, XGBoost, GitHub Actions, AWS ECR/EC2) - Designed an end-to-end MLOps pipeline with modular architecture, implementing automated CI/CD workflows that reduced development time by 30%. - Engineered scalable ETL processes and predictive analytics system using MLflow experiment tracking, achieving 15% improvement in F1-score for churn prediction. - Developed and deployed containerized prediction API on AWS EC2/ECR through GitHub Actions, enabling robust model serving with automated testing and versioning.	[Link]

SKILLS

Languages	Python, R, SQL, Java, JavaScript/TypeScript, C++, Go
Databases	PostgreSQL, MySQL, Redshift, MongoDB, Neo4j, GraphQL, Cassandra, Vector Databases
Libraries	React, TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, Django, DSPy, JAX, Langfuse
Technologies/Tools	Git, AWS, GCP, Azure, JIRA, Docker, TDD, GPU Optimization
Skills	Front-end, Back-end, Full-Stack, Data Engineering, Microservices, MLOps, LLMOps
Machine Learning	Deep Learning, Natural Language Processing, Reinforcement Learning, Computer Vision

EDUCATION

MS in Statistics and Data Science, University of Central Florida (UCF) (GPA: 4.00) Coursework: Neural Networks, Data Science & Analytics, Software Engineering for Data Science, Applied Statistics	08/2025 - Present
Bachelors in Computer Science and Engineering (Data Science), Osmania University (GPA: 3.66) Coursework: Artificial Intelligence, Machine Learning, Data Mining, Algorithms	07/2020 - 07/2024

ACHIEVEMENTS

Advanced Certification in AI & ML - IIIT-Hyderabad	04/2025 - 11/2025
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